

increase the sensing threshold of the leadless pacemaker to suppress oversensing

Executive Technology Brief

Domain: rf_wireless

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Pacemaker battery depletion requiring surgical replacement

Mechanism: mechanically mediated device-device interaction causing intermittent oversensing

Architecture: PROPOSED architecture (no physical system exists): transmit chain + power limits; channel / tissue path (path loss model); receive chain + detection; regulatory exposure constraint (SAR) — realizing: increase the sensing threshold of the leadless pacemaker to suppress oversensing

Governing Model: RF / wireless link governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

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{'equation_id': 'RF-001', 'name': 'Friis link budget', 'expression': 'P_r = P_t * G_t * G_r * (lambda / (4 * pi * d))^2', 'source': {'text': 'Friis transmission equation (standard antenna-theory relation)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'far field, line of sight; tissue path requires additional loss term', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['in-body path loss exceeds free-space – margin is MODEL_DERIVED'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions': ['in-body path loss exceeds free-space – margin is MODEL_DERIVED'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': [], 'model_applicability': 'far field, line of sight; tissue path requires additional loss term', 'model_assumptions': ['in-body path loss exceeds free-space – margin is MODEL_DERIVED'], 'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
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{'equation_id': 'RF-003', 'name': 'Thermal noise floor', 'expression': 'N = k * T * B', 'source': {'text': 'standard thermal-noise relation (communications texts)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'matched, linear receiver at physical temperature T', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['noise figure and interference add on top of the thermal floor and must be budgeted explicitly'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions': ['noise figure and interference add on top of the thermal floor and must be budgeted explicitly'], 'violations': [], 'epistemic_class': 'MODEL_DERIVED (mechanically evaluated, recorded for audit)'}, 'mechanism_mentions': [], 'model_applicability': 'matched, linear receiver at physical temperature T', 'model_assumptions': ['noise figure and interference add on top of the thermal floor and must be budgeted explicitly'], 'invention_tie': {'linkage_kind': 'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
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{'equation_id': 'RF-002', 'name': 'Signal-to-noise ratio', 'expression': 'SNR = P_signal / P_noise', 'source': {'text': 'standard detection-theory quantity (communications texts)', 'epistemic_class': 'EXTERNAL_PRECEDENT', 'verify_before_release': True}, 'applicability': {'condition': 'defined for the chosen detection bandwidth', 'epistemic_class': 'MODEL_DERIVED', 'judged_for_domain': 'rf_wireless'}, 'assumptions': ['noise figure and interference terms must be budgeted explicitly'], 'selection_rationale': {'epistemic_class': 'MODEL_DERIVED', 'verdict': 'CONDITIONAL', 'reason': 'domain template justifies this relation for the domain, but the invention text does not yet engage its variables; applicability must be
```

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confirmed by the first characterization', 'engaged_variables': [], 'assumption_check': {'assumptions':  
['noise figure and interference terms must be budgeted explicitly'], 'violations': [], 'epistemic_class':  
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and interference terms must be budgeted explicitly'], 'invention_tie': {'linkage_kind':  
'applicability_judgment', 'verdict': 'CONDITIONAL', 'engaged_variables': [], 'mechanism_mention_count': 0}}
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Kill Condition: killer experiment fails: falsification_test_from_candidate (pre-registered pass/fail required before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.